American Computer Science League

2024 Finals ● Short Problems ● Junior Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$AB + \overline{A} + B + \overline{A}\,\overline{B}$$

A. $\overline{A} + \overline{B}$
B. $\overline{A} + B$
C. $A + \overline{B}$
D. $A + B$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$\overline{A(\overline{B} + C)} + B\overline{(A + C)} + \overline{C}$$

A. 2
B. 3
C. 4
D. 5
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

(RCIRC-3 01100) OR (NOT (10110 AND 10011))

A. 10001
B. 01101
C. 10010
D. 11101
E. None of the above

4. Bit-String Flicking

Solve for X, a bit-string of 5 bits:

(LCIRC-2 (10111 AND X)) = (RSHIFT-1 10100)

A. 01*10
B. 1*010
C. 111*0
D. 1100*
E. None of the above

5. Recursive Functions

Find $f(f(18))$, given:

$$f(x) = \begin{cases} f(x-1) + 1 & \text{if } x \text{ is a multiple of } 2 \text{ and } 3 \\ f(x+2) - 2 & \text{if } x \text{ is a prime} > 5 \\ x - 7 & \text{otherwise} \end{cases}$$

A. 4
B. 11
C. 15
D. 21111
E. None of the above

6. Recursive Functions

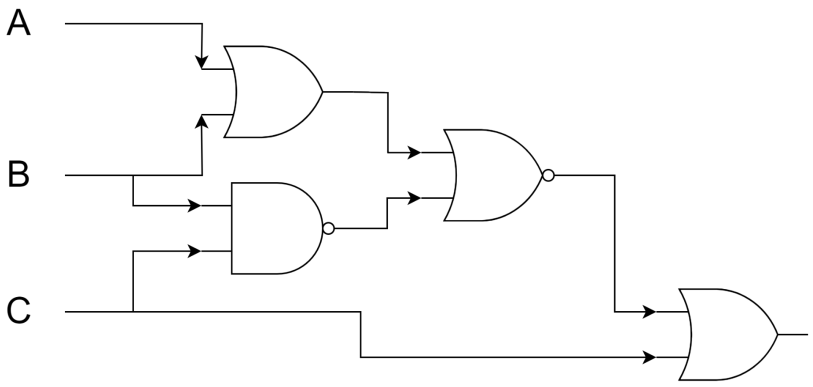
Find $f(13, 7)$, given:

$$f(x,y) = \begin{cases} f(x-1, y+2) + 1 & if\ x > y \\ f(x-1, y+1) - 3 & if\ x = y \\ y - x & if\ x < y \end{cases}$$

A. -1
B. 0
C. 1
D. 2
E. None of the above

7. Digital Electronics

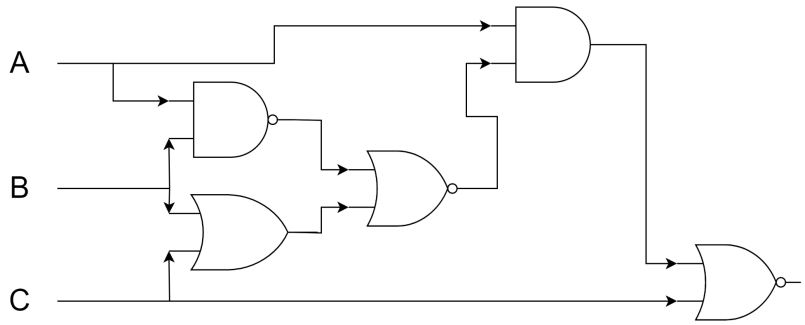
Simplify the Boolean expression that corresponds to the following digital circuit:



A. A
B. $B\,C$
C. C
D. $\overline{A}\,C$
E. None of the above

8. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. None of the above

9. Prefix-Infix-Postfix

In the following prefix expression, all numbers are single digits. Translate it into postfix.

+ − * 2 − / * 6 2 3 1 ^ 2 3 5

A. 2 6 * 2 3 / 1 −*2 3^−5+
B. 2 6 2 * 3 / 1 −*2 3^−5+
C. 2 6 2 * 3 1 / −*2 3^−5+
D. 2 6 2 * 3 / 1 −*2 3^5 −+
E. None of the above

10. Prefix-Infix-Postfix Define: $a \# b = a^2 - b$ Evaluate the following postfix expression if all numbers are single digits: 4 2 # 7 / 1 1 # 2 * − 5 9 # +	A. 0 B. 2 C. 16 D. 18 E. None of the above
11. Computer Number System Convert 155414_{16} to octal.	A. 5525204 B. 5252024 C. 5272024 D. 5242024 E. None of the above
12. Computer Number Systems Evaluate the following and express the result in hexadecimal: $AAAA_{16} + BBB_{16} + CC_{16} + D_{16}$	A. A74B B. B73E C. B64D D. C72C E. None of the above
13. Data Structures Build a binary search tree for: **MICROWAVEOVEN** What is the internal path length?	A. 26 B. 30 C. 32 D. 45 E. None of the above

14. Data Structures

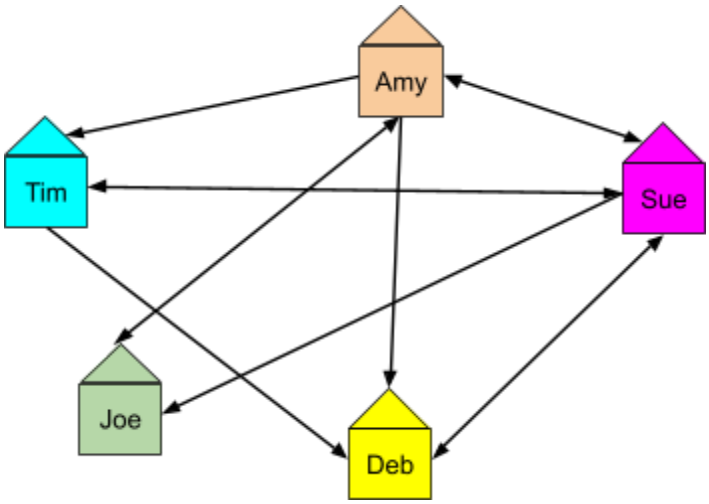
Consider a rack container that holds basketballs vertically. You can put a basketball at the top of the rack, and remove the basketball at the top. Initially, there are 5 basketballs on this stack, numbered 1-5 from top to bottom. Each POP stores the number of the basketball given to that student, identified as A-K. Every PUSH places that student's numbered ball back onto the top of the stack. What students use ball 2?

```
A = POP()
B = POP()
C = POP()
D = POP()
PUSH(B)
E=POP()
PUSH(C)
PUSH(A)
F=POP()
G=POP()
PUSH(F)
H=POP()
PUSH(D)
J=POP()
PUSH(E)
K=POP()
```

A. B, F, G, K
B. B, E, K
C. B, E, G, K
D. C, E, K
E. None of the above

15. Graph Theory

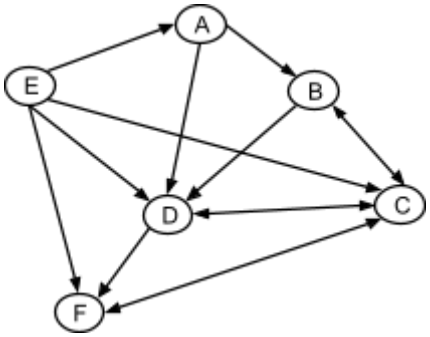
The graph below shows paths between friends' houses. Which friends can visit all other friends without visiting a friend more than once or going back to their own house until the very end?



A. All but Joe
B. Only Amy
C. Tim, but not Deb
D. All of them
E. None of them

16. Graph Theory

How many cycles are in the graph below?



A. 4
B. 5
C. 6
D. 7
E. None of the above

17. What Does This Program Do? (Branching)

What needs to replace the <*condition*> and <*statement*> in the following code so that the 3 integers are output in descending order?

```
input a, b, c
if a < b then
      swap(a,b)
end if
if b < c then
     swap(b,c)
end if
if <condition> then
    <statement>
end if
output a, b, c
```

A. a < c, swap(a,c)
B. a < b, swap(a,b)
C. a > b, swap(a,b)
D. b < c, swap(b,c)
E. None of the above

18. What Does This Program Do? (Loops)

What is the output when the following program is executed?

```
n = 2
val = 2^n * (2^(n+1)−1)
sum1 = 0
for y = 1 to 2^(n+1)−1
     sum1 = sum1 + y
next y
sum2 = 0
for y = 1 to 2*n−1 step 2
     sum2 = sum2 + y^3
next y
output val − sum1 + sum2
```

A. 0
B. 28
C. -28
D. 56
E. None of the above

19. What Does This Program Do? (Arrays)

Variables **num** and **arr** are arrays containing integers. Given that array **arr** contains the 10 values below, what is output when the following program is run?

4	3	2	1	3	4	2	3	4	3

```
for x = 1 to 4
     num(x-1) = 0
next x
for x = 0 to 9
     num(arr(x)-1) = num(arr(x)-1) + 1
next x
m = 0
for x = 1 to 3
     if num(x) > num(m) then
          m = x
     end if
next x
output num(m)
```

A. 1
B. 2
C. 3
D. 4
E. None of the above

20. What Does This Program Do? (Strings)

What is the output for the following program?

```
ans = ""
s = "10101 01001"
left = s[:5]
right = s[5:]
for x = 4 to 0 step -1
     if left[x] == right[x] then
          ans = "0" + ans
     else
          ans = "1" + ans
     end if
next x
output ans
```

A. 11100
B. 00011
C. 11010
D. 00101
E. None of the above